HierU-Net: A Hierarchical Semantic Segmentation Method for Land Cover Mapping

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***Abstract*— Land cover mapping is crucial for natural resource assessment, urban planning, and sustainable development. Land cover nomenclature often includes two or three hierarchical levels with tree-like hierarchical structures. This study aims to explore these hierarchical relationships and the potential of hierarchical semantic segmentation for land cover mapping. We propose a hierarchical semantic segmentation architecture by taking advantage of dual U-shaped network, named as HierU-Net. The coarse-level result is ingested to the fine-level segmentation functioned as soft constraints. The propagation of error will not be certain. Moreover, we employ a multitask loss function weighted by homoscedastic uncertainty to optimize the training. To evaluate the performance of the proposed method, we create a hierarchical semantic segmentation dataset (HierToulouse), which contains 11 528 samples, including images and land cover labels at two hierarchical levels. The experiments demonstrate that the proposed approach is capable of achieving accurate land cover segmentation at both coarse and fine levels, with segmentation results surpassing those obtained using the flat method.**

Manuscript received 25 September 2023; revised 23 January 2024; accepted 2 March 2024. Date of publication 18 March 2024; date of current version 26 March 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 42271482, Grant 42101070, and Grant U23A2020; in part by the Knowledge Innovation Program of Wuhan-Shuguang Project under Grant 2022010801020284; in part by the International Science and Technology Cooperation Program of Hubei Province under Grant 2022EHB019; in part by the Macao Young Scholars Program under Grant AM2022013; in part by the Science and Technology Development Fund of Macau under Grant 0059/2020/A2 and Grant 0052/2020/AFJ; in part by the Zhuhai Industry-University-Research Collaboration Program under Grant ZH22017002210011PWC; and in part by the Fundamental Research Funds for the Central Universities under Grant CCNU22QN018. *(Corresponding author: Zhanchuan Cai.)*

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Data is available on-line at https://geoservices.ign.fr/.

Digital Object Identifier 10.1109/TGRS.2024.3376432

***Index Terms*— Hierarchical semantic segmentation, land cover mapping, U-Net, very high-resolution (VHR) imagery.**

I. Introduction

LAND cover information plays a vital role in various fields such as urban planning, environmental monitoring, and natural resource management [1], [2], [3], [4], [5]. The advent of remote sensing technology has made it easier to obtain large-scale data conveniently, maintaining continuous spatial coverage with a short revisit period. It acts as an effective data source for land cover mapping and change monitoring [6], [7]. With the rapid development of sensor technology, a large number of very high-resolution (VHR) remote sensing images have been commercialized, which provides new opportunities for extracting detailed land cover information [8].

Land cover mapping using remotely sensed imagery is traditionally implemented through different approaches: pixel-based, object-based, and scene-based classification [9]. The pixel-based approach is easily influenced by salt-and-pepper noise effects, and the rich spatial information of VHR data cannot be fully utilized [10], [11]. Object-based and scene-based classification methods need to segment the image to group pixels into an object or scene. Image segmentation is a difficult task, which needs to consider many factors, such as target size, shape, and color. Especially, it is not easy to determine the segmentation scale. The classification accuracy is often affected by the segmentation result [12].

In recent years, deep learning methods, especially convolutional neural networks (CNNs) [13], have shown excellent performance in various computer vision tasks [14]. CNNs are a type of neural networks that are suitable for processing image data. They consist of multiple interconnected layers, which are utilized to extract features from the input image. The ability of CNNs to learn features from input images automatically has made them a popular choice for tasks such as land cover classification, vegetation analysis, urban development monitoring, and disaster assessment [15], [16], [17]. Seydi et al. [18] proposed a novel end-to-end multidimensional CNN framework for monitoring land cover changes using multisource remote sensing datasets, and multiple parallel channels are utilized to extract deep features.

Fully connected networks (FCNs) and encoder–decoder-based networks, such as SegNet [19] and U-Net [20], could achieve high accuracy in pixel-by-pixel image segmentation.

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Although the model structure and experimental setup are different, they perform well in the task of land cover classification. Ghosh et al. [21] proposed the stacked U-Nets, which allows the details captured at high resolution to be combined with the long-distance context information captured at low resolution for the purpose of generating a segmentation map of the scene. Ji et al. [22] proposed a domain-adaptive land cover classification method based on generative adversarial network (GAN). Recently, a novel GAN-based single-source domain expansion network (SDEnet) is developed by using only source domain for model training [23]. Domain adaption (DA) is a challenging task that makes the model from source domain generalize into target domain dealing with the discrepancy across domain distributions. Methods including Topological structure and Semantic information Transfer network (TSTnet) [24] and multilevel DA network (MDA-NET) [25] are also proposed to deal with the challenge. Such techniques are promising in land cover mapping using remotely sensed data with various spatial, temporal, and spectral characters.

Land cover mapping using remotely sensed imagery is commonly treated as generic image classification or semantic segmentation tasks following a flat way, in which all land cover classes are classified simultaneously in a "one-against-all" approach [26]. However, most national or international land cover nomenclatures adopt a hierarchical tree-like structure. For example, CORINE land cover (CLC) has three hierarchical levels, with a total of 44 land cover classes. At the first level, it includes five main categories like *Artificial surfaces*, *Agricultural areas*, *Forest and seminatural areas*, *Wetlands* and *Water bodies*. At the second level, 15 types of land cover are distinguished. The third level covers the most detailed land cover classes including *Continuous urban fabric*, *Port areas*, and *Dump sites*. Using hierarchical information in land cover mapping is helpful to better reflect the internal relationship and dependence between different land cover categories. The classification decision is not only influenced by the adjacent categories, but also by the broader land cover categories to which it belongs.

The land cover hierarchical information encoded in the tree-like structure has been explored in several related studies [27], [28], [29]. The hierarchical classification is performed in a sequential and hard-coding way, which means labels predicted for finer levels have to be children of the categories predicted for the coarser levels of the hierarchy. Considering that the error will be propagated if the prediction at fine level is incorrect, some experiments suggest that hard-coding hierarchical classification methods may not yield superior results compared to flat classification [30]. There are several deep learning-based methods exploring the hierarchical information encoded in land cover nomenclatures. Yang et al. [31] proposed a CNN-based framework with joint optimization for hierarchical land use and land cover classification. Interacting classification layers were considered to predict one label per level and integrate the semantic dependencies between the classes at different levels. The inference was operated at the object level. Li et al. [32] proposed a convolutional recursive neural network based on hierarchical category structure, which constructs the dependency relationship between different layers from top to bottom through the leaf nodes of the input layer based on hierarchical structure. The results showed that it can effectively classify land cover classification and has the potential to accurately map land cover at a large scale.

For semantic segmentation of remotely sensed imagery, the hierarchy primarily refers to different feature levels, such as the hierarchical fusion of features from different stages and scales, aiming to refine feature maps and enhance the model's capability to recognize various types of ground objects [33]. The complex multilevel nomenclature for land cover semantic segmentation has not been investigated yet. Progress has been made in street scene understanding and human parsing recently [34], [35], [36], [37]. Li et al. [34] devised a hierarchical semantic segmentation framework enforcing segmentation predictions to obey the hierarchy structure during network training, and pointed out that the task is still largely unexplored in the current literature. The research assumption is that incorporating the structure information of land cover nomenclatures into the semantic segmentation model will improve the performance compared to solely flat methods.

In this research, we present a hierarchical semantic segmentation framework taking advantage of the nomenclature information. A dual U-shaped network is proposed to hierarchically identify land cover types at coarse level and fine level. Several backbones including MobileNet, EfficientNet, VGG19, ResNet50, and DenseNet-121 were evaluated and the last one was chosen as the encoder of the network [38], [39], [40], [41], [42]. The main contributions are stated as follows.

1) We design a hierarchical semantic segmentation framework for multilevel land cover mapping using an encoder–decoder network structure. It is able to classify land cover at coarse and fine levels hierarchically. To avoid the propagation of error, the coarse-level result is ingested to the fine-level segmentation functioned as soft constraints.
2) We employ a multitask loss function weighted by homoscedastic uncertainty to optimize the training task. The losses from the coarse and fine semantic segmentation tasks can be automatically adjusted, balancing the performance across the two levels of land cover categories.
3) A large-scale dataset (that we called HierToulouse) is created from VHR imagery and land cover labels corresponding to two hierarchical levels. The dataset is considered for the task of hierarchical semantic segmentation of land cover mapping. This is a data-oriented contribution that will allow the reproducibility of our approach and open new research since well-annotated land cover datasets with hierarchical semantic labels are difficult to obtain, and no such datasets are currently publicly available. The dataset can be downloaded at https://zenodo.org/records/10532778.

The article is organized as follows. Section II introduces the HierToulouse dataset including VHR imagery and hierarchical land cover labels. Section III describes the proposed hierarchical semantic segmentation model and evaluation metrics. In Section IV, the experimental results and consistency

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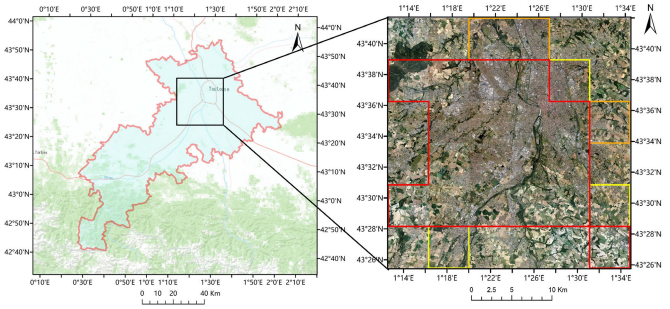


Fig. 1. Location of the study area. The region within the red box is considered for training, and within orange and yellow boxes used for validation and testing, respectively.

analysis are shown. The conclusion of this article is given in Section V.

II. Dataset

The chosen study area is Toulouse (see Fig. 1), the fourth largest city in France located in the south of France. It is one of the important aviation and aerospace centers in Europe with latitudes from 1°14′ E to 1°32′ E and longitudes from 43°36′ N to 43°57′ N, as shown in Fig. 1. Toulouse has complex topography, including rivers, mountains, plains, and man-made structures, and is rich in natural and human landscape resources.

Although there are a variety of remotely sensed datasets publicly available for land cover classification, few of them is suitable for the task of hierarchical semantic segmentation. This study area was chosen since it was the first region where the French National Institute of Geography and Forest Information (IGN) LULC data produced a new LULC object-based product starting with 2013. Each polygon is classified according to its land cover class (e.g., built-up area) and land use class (e.g., residential). Three time snapshots exist—2013, 2016, and 2019—and they are published with an open data license. The nomenclature of this LULC database is hierarchically organized for both land cover (four levels) and land use (three levels) layers.

A. Remotely Sensed Data

Remotely sensed imagery from the French national aerial orthoimage database BD ORTHO provided by IGN is used. Images used in this study were acquired in 2019 and consist of three spectral bands including red (R), green (G), and blue (B) with a ground sampling distance of 0.5 m.

B. LULC Label Data

The land cover labeling data is a part of IGN's OCS-GE database. OCS-GE is an LULC database consisting of two distinct sets of layers for LU and LC. It is mostly generated automatically out of several existing databases, and then completed and corrected though a manual labeling after visual interpretation. For those that are difficult to identify directly based on remote sensing images, accurate land cover types are obtained through field survey. Land cover in the study area has four primary or coarse classes including *Artificial surface*, *Natural surface*, *Woody vegetation*, and *Nonwoody vegetation*. For secondary or fine classes, there are *Construction area*, *Nonconstruction area*, *Mineral material area*, *Water surface*, *Broad-leaved forest*, *Coniferous forest*, *Mixed forest*, *Shrubbery*, *Other woodlands*, and *Herbaceous plants*. The hierarchical relationship between the coarse-level category and the fine-level category is shown in Fig. 2.

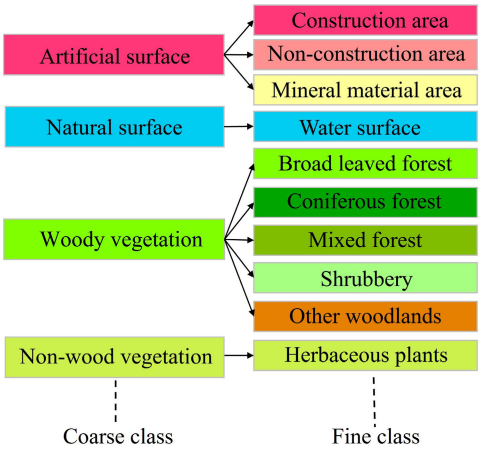


Fig. 2. Hierarchical relationship between the coarse-level category and the fine-level category.

It must also be said that the geometry of OCS-GE is supposed to respect some generalization constraints defined formally in the specifications of the product for cartographic purposes. For example, mixed natural/vegetated areas will be classified with the primary cover of *Natural surface* if the absolute coverage of woody or nonwoody vegetation is less than 25%. Additionally, herbaceous areas (garden, lawn) nearby buildings could be merged with the adjacent polygon. This can lead to discrepancies between OCS-GE and VHR imagery (e.g., merged groups of buildings for the *Construction area* class).

C. Construction of Training Data

Starting from the input data described previously, we build a suitable dataset for land covering mapping using hierarchical semantic segmentation, named as HierToulouse.

First, we converted land cover layer coming from the OCS-GE vector data into two raster layers representing coarse-level and fine-level land cover category. Thus, each pixel having 0.5 meters resolution is characterized by values from both the second (coarse) and the most detailed (fine) land cover code.

Then, VHR data with paired coarse-level and fine-level land cover labels were divided into 512 × 512 patches. Each patch was generated using a sliding window across the complete image with a stride of 0.2. This method ensured that all edge pixels within each patch could also serve as center pixels for subsequent patches. In total, 11528 paired image patches are generated in the study area.

The complete dataset was subsequently split into training, validation, and test datasets. Specifically, 8624 images were allocated for training purposes, while 1452 images were designated for validation, and an additional 1452 images were set aside for evaluating performance during testing. Examples of training data are shown in Fig. 3.

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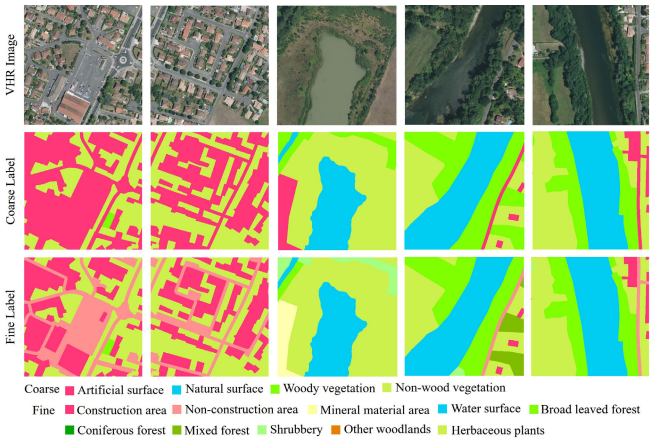


Fig. 3. Examples of the VHR image, coarse label, and fine label included in the training set.

III. Methodology

A. U-Net and Backbones

The original U-Net architecture was introduced in [20], which achieved well on cell segmentation in biomedical images with a small amount of training data. The distinctive feature of U-Net is the combination of the encoder and decoder parts of a convolutional neural network, enabling the capture of contextual information at multiple scales and its propagation to higher resolution layers, thanks to skip connections. The encoder is responsible for gradually extracting feature information from the image, while the decoder maps these features back to the original image dimensions, generating pixel-level segmentation results. The introduction of skip connections in U-Net facilitates the transfer and fusion of information between the encoder and decoder, thereby enhancing the segmentation precision. Due to the success of U-Net, recent works in semantic segmentation [19], [43] and object detection [44] have also proposed based on encoder–decoder architectures.

In the field of deep learning, "backbones" refers to the basic module or backbone structure in the neural network architecture, which is used to extract the features of input data. Backbones play a key role in the deep learning model. They can learn the low-level and high-level features of input data, providing meaningful representation for the model. Common backbones include the following.

1) VGG: VGG is a classical convolutional neural network with multiple convolution layers and pooling layers. Its structure is relatively simple, but it performs well in some tasks [40].
2) ResNet: ResNet introduces residual blocks, which allows the network to be deeper and effectively solves the problem of gradient disappearance. It has variants with different depths, such as ResNet-50 and ResNet-101 [41].
3) DenseNet: DenseNet adopts dense connections. Within each dense block, the outputs of all previous layers are connected at each layer. This structure is helpful for information flow and gradient propagation [42].
4) EfficientNet: EfficientNet can improve the performance and efficiency by adjusting the depth, width, and resolution of the network at the same time, which is a popular choice in the case of limited computing resources [39].
5) MobileNet: MobileNet is designed to be lightweight and efficient for mobile and embedded vision applications. It employs depthwise separable convolutions, reducing computational complexity while maintaining competitive performance [38].

These backbones could be integrated with U-Net or other image segmentation architectures to extract image features, so as to achieve accurate and robust segmentation results.

B. The HierU-Net Architecture

This section describes the proposed HierU-Net architecture considering two U-shaped networks via joint training. Fig. 4 shows an overview of our proposed architecture. DenseNet-121 is used as the encoder, and then the feature map is processed, followed by the decoder. The difference between the U-shaped network in HierU-Net and the traditional U-Net lies in the use of the pretrained DenseNet-121 as the encoder and the addition of an Atrous Spatial Pyramid Pooling (ASPP) module [45]. Each convolution layer in DenseNet is connected with the outputs of all previous layers, thus realizing the sharing of features without redundant convolutions. This structure reduces the number of parameters in the network, and can effectively solve the problem of gradient disappearance and gradient explosion. Features of different scales can be effectively combined improving the performance of the model [42].

In the first network module (Network-1), the input image patch is fed into a modified U-shaped network, which generates a mask for the prediction of coarse-level classes. The original image patch concatenated with the generated coarse-level land cover mask is used as the input of the following network module (Network-2), generating the fine-level mask for land cover classification. The two U-shaped networks have basically the same structure, and the structure of a single U-shaped network is shown in Fig. 5. By employing two U-shaped network architectures sequentially, the proposed HierU-Net is able to classify coarse-level and fine-level classes of land cover simultaneously. The semantic segmentation output at the coarse level acts as soft constraints to the fine-level land cover segmentation, expecting to achieve better results for land cover mapping.

C. Encoder Explanation

For each U-shaped network, each encoder tries to encode the information contained in the input image. Both encoders using DenseNet-121 are pretrained on ImageNet dataset [46]. In the first stage of the encoder, DenseNet-121 uses a 7×7 convolution layer to extract features, and the number of output feature maps is 64. The difference between Encoder1 and Encoder2 is that the kernel sizes in the convolution layer here are $7 \times 7 \times 3$ and $7 \times 7 \times 4$, respectively, and the rest of the structures are the same [47]. In the second stage, the feature map is processed by six dense layers, where the input for each convolution layer consists of the feature map concatenated with the outputs of all previous convolution layers within that block. Each dense layer uses a 1×1 convolution kernel and

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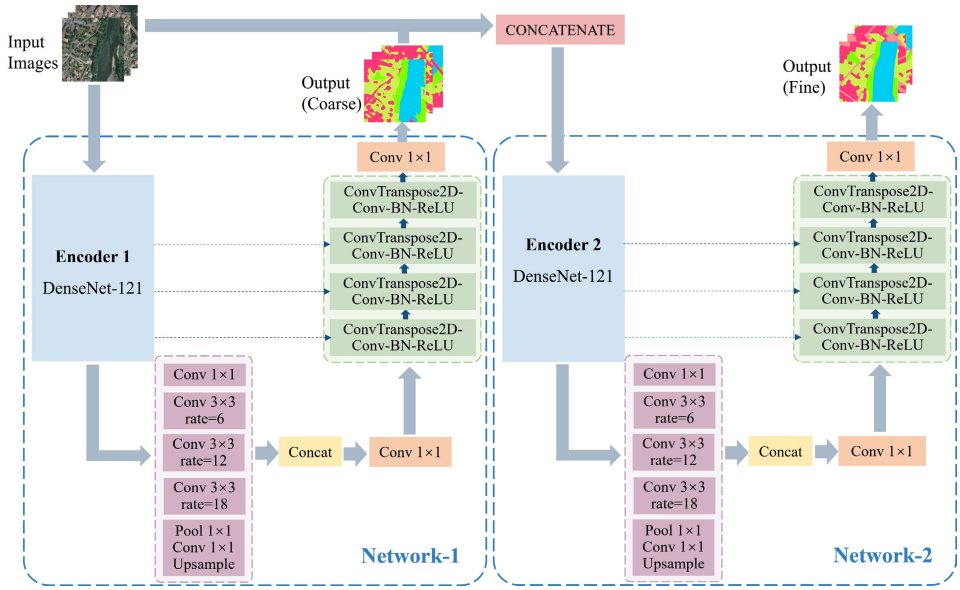


Fig. 4. Architecture diagram of the proposed HierU-Net. On the left is the network module for coarse-level segmentation, and on the right is for fine-level segmentation. Each network consists of an encoder, a decoder, and an ASPP module.

a 3 × 3 convolution kernel for convolution operation, and the number of output feature maps is 32. Therefore, the number of final output feature maps of this layer is $64 + 32 \times 6 = 256$. In the third stage, the feature map is processed by a transition layer. In order to reduce the dimension of the feature map, a 1 × 1 convolution kernel is used for convolution operation, and a 2 × 2 average pooling operation is carried out, so that the size of the feature map is reduced to a quarter of the original size. The number of output feature maps is 128. Similarly, after 12 dense layers, 1 transition layer, 24 dense layers, 1 transition layer, and 16 dense layers, the number of output feature maps is 512, 256, 1024, 512, and 1024, respectively.

D. Feature Map Processing

An ASPP module is considered between the encoder and decoder to enhance the performance of the proposed model. Fig. 6 illustrates the schematic of the ASPP module, which utilizes a multiscale receptive field mechanism to capture rich feature information at different scales. By sampling feature maps in parallel with holes of varying sampling rates [48], the ASPP module fuses multiscale features to expand the number of channels. Then, the number of channels is reduced to the expected value via a 1 × 1 convolution operation, enhancing the receptive field of the model and further improving the performance [49].

E. Decoder Explanation

Deconvolution method is used to up-sample the feature map, and the weight of each convolution kernel is learned in the deconvolution process. This operation disperses each element of the input feature map into a larger space by performing convolution operation at each position in the feature map to restore the details and features of the original image. In addition, in the decoder part, skip connection is used to combine the encoder feature map with the decoder feature map, so that the network can pay attention to local and global features at different scales at the same time, thus better preserving details and improving the accuracy of segmentation.

F. Loss Function

When dealing with multitask training, the model could be optimized by a joint loss function assigning different weight values to the loss of each task [50]. It is helpful to handle the hierarchical relationship between two tasks and improving the performance. The cross-entropy loss function is considered for coarse-level and fine-level training in the present study. Cross-entropy loss function is an effective loss function to measure the difference between the prediction and ground truth of segmentation. The calculation formula is denoted as follows:

$$L = -\frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{M} y_{ij}\log\left(p_{ij}\right) \tag{1}$$

where n is the number of samples, M is the number of classes, y_{ij} is the binary indicator (0 or 1), and p_{ij} is the predicted probability value.

Homoscedastic uncertainty has been shown as an effective way to weight losses [51]. It is considered when designing the multitask loss function to balance hierarchical tasks during training. The idea is to assign the loss value according to the complexity of the task and making weights as trainable parameters, which will be updated by the optimizer during the training process. For each task, the model adaptively adjusts its weight so that the training task will be optimized. Specifically, for the two semantic segmentation tasks as described in the proposed HierU-Net, the loss values are represented using cross-entropy loss functions, namely $L1$ for coarse-level segmentation and $L2$ for fine-level segmentation. The overall loss is a weighted sum of $L1$ and $L2$. The loss function is denoted as follows:

$$L(W, \sigma_1, \sigma_2) = \frac{1}{2\sigma_1^2}L_1(W) + \frac{1}{2\sigma_2^2}L_2(W) + \log\sigma_1\sigma_2 \tag{2}$$

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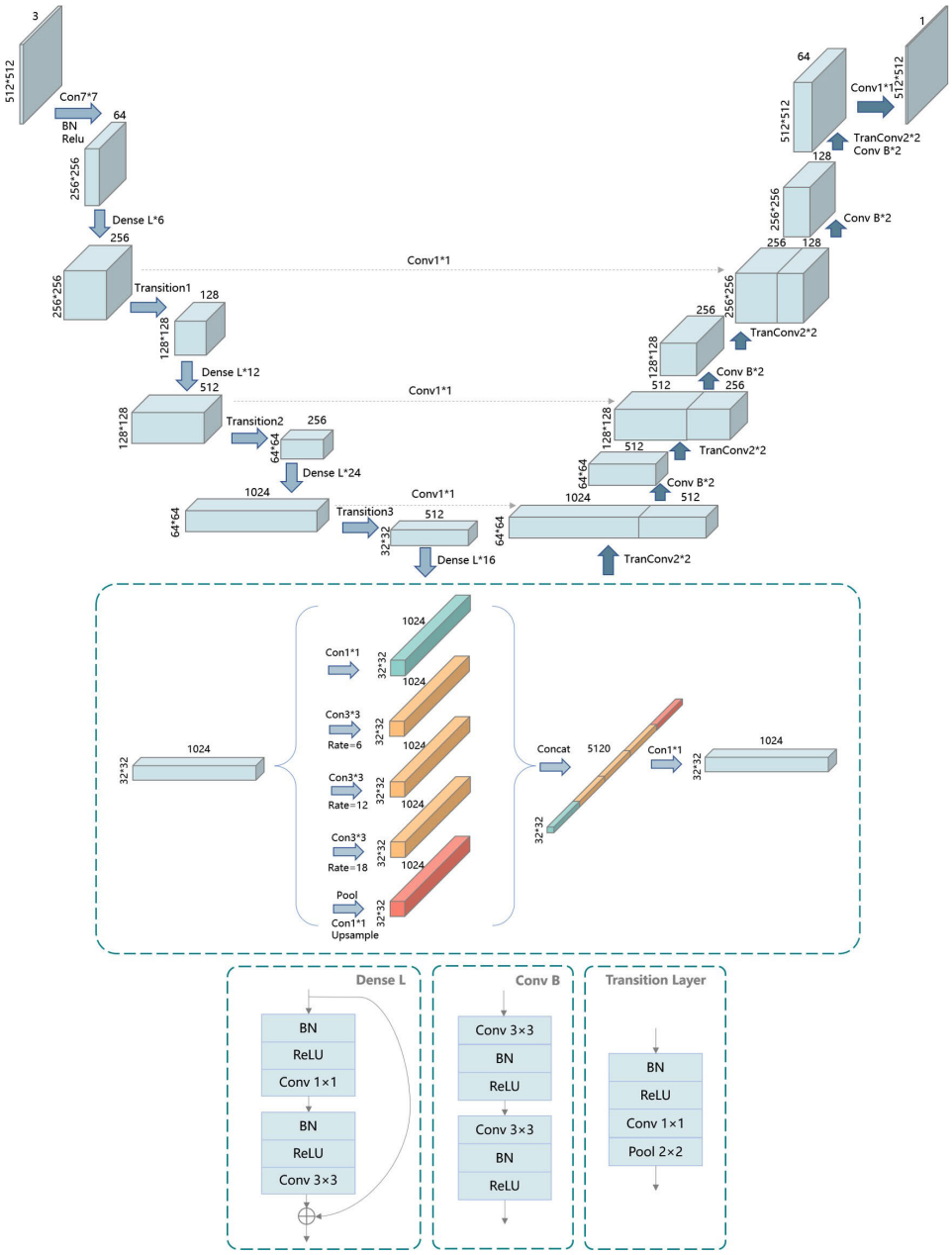


Fig. 5. Diagram of a single U-shaped network, adapted from U-Net, incorporating dense block modules and ASPP module.

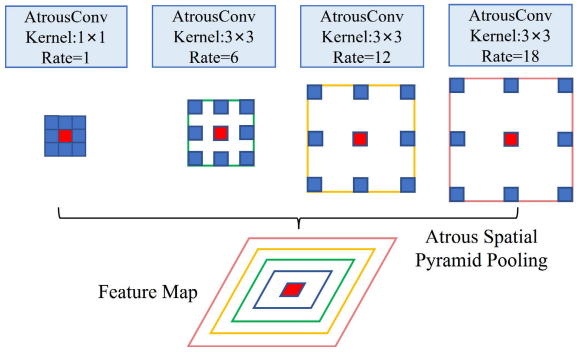


Fig. 6. Schematic of ASPP module.

where $L1$ is the loss of the coarse-level task, $L2$ is the loss of the fine-level task, and σ_1 and σ_2 are the weights of the two losses.

G. Experimental Setting

Experiments were performed on VHR orthorectified imagery and land cover labels from OCS-GE database in Toulouse, France. For flat semantic segmentation using U-Net, the effects of different backbones were evaluated including MobileNet, EfficientNet, VGG19, ResNet50, and DenseNet-121. The backbone with the best performance was selected for HierU-Net to classify coarse-level and fine-level land cover information simultaneously. The performance between flat and hierarchical segmentation for land cover was compared.

During the model training process, the stochastic gradient descent (SGD) algorithm with momentum was employed for optimization, continuously adjusting parameters to minimize the loss function. The learning rate is set to 0.0001, the number of epochs is set to 50, and the batch size is 4. Both training and testing processes are implemented using the PyTorch

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framework, and the model is trained on an RTX4090 with 24GB GPU memory.

H. Evaluation Metrics

Quantitative evaluation is an important step to assess the performance of the proposed method. A confusion matrix is computed based on ground truth data and predictions. The confusion matrix is easy to create and interpret, providing rich information in terms of average accuracy of the whole dataset and also on individual classes. Several metrics could be computed from the confusion matrix, including overall accuracy (OA), user's accuracy (UA), producer's accuracy (PA), IoU, mIoU, and frequency-weighted intersection over union (FWIoU). IoU measures the overlap between predicted and ground truth regions. MIoU calculates the average IoU across all classes, providing insights into the model's ability to accurately segment multiple classes simultaneously. FWIoU takes into account both per-class IoU and class frequency [52]. The specific definitions are as follows:

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{3}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{4}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{5}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{6}$$

$$\text{mIoU} = \frac{\frac{\text{TP}}{\text{TP}+\text{FP}+\text{FN}} + \frac{\text{TN}}{\text{TN}+\text{FN}+\text{FP}}}{2} \tag{7}$$

$$\text{FWIoU} = \left[\frac{\text{TP} + \text{FN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}}\right] \times \left[\frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}\right] \tag{8}$$

where TP, FP, FN, and TN refer to the number of true positives, false positives, false negatives, and true negatives, respectively.

IV. Results and Discussion

In this section, we first present the results obtained using the flat U-Net model in our study area. We then compare HierU-Net models with other semantic segmentation methods. Subsequently, we analyze the segmentation consistency between coarse and fine categories. Visual interpretation is provided through saliency maps to better understand what representations the hierarchical model learned from the inputs. Ablation experiments are conducted, and then the performance of HierU-Net is further validated on the GID dataset.

A. Land Cover Semantic Segmentation Using U-Net With Different Backbones

In the flat experiment, U-Net with different backbones including MobileNet, EfficientNet, VGG19, ResNet50, and DenseNet-121 are explored on the HierToulouse dataset. Table I displays the performance of each model on coarse and fine levels. The segmentation results are much better at the coarse level, which have only four land cover categories. Models that employed VGG19, ResNet50, and DenseNet-121 as backbones show better performance in both coarse and fine classes. Among them, the DenseNet model has the highest OA value, achieving 89.22% and 86.05% in coarse and fine classes, respectively.

TABLE I
Performance Comparison of Flat Semantic Using Different Backbones

Model	Level	OA	PA	UA	MIoU	FWIoU
U-Net	Coarse	87.86	79.80	87.09	71.05	78.47
	Fine	83.77	42.79	70.29	36.01	73.96
Mobile-UNet	Coarse	84.33	69.42	85.73	60.75	72.89
	Fine	79.91	32.92	65.15	27.71	66.44
Efficient-UNet	Coarse	84.96	70.31	86.02	61.29	73.92
	Fine	80.52	33.99	74.52	28.01	67.65
VGG-UNet	Coarse	88.03	**84.90**	87.20	75.31	80.37
	Fine	84.03	**46.99**	75.09	37.05	74.98
Res-UNet	Coarse	88.84	84.80	87.66	75.82	80.09
	Fine	85.55	43.70	76.02	**37.25**	75.13
Dense-UNet	Coarse	**89.22**	84.02	**88.87**	**76.05**	**80.61**
	Fine	**86.05**	42.43	**77.06**	37.06	**75.58**

The performance on each fine class is presented in Table II. *Mixed forest*, *Coniferous forest*, and *Other woodlands* have a very small proportion across the whole dataset and are nonuniformly distributed. Especially, pixels belonging to the class of *Other woodlands* in the test dataset are absent. Thus, these three classes are not considered for the analysis. From Table II it can be seen that each model performed well on classes dominating within the study area, such as *Construction* and *Herbage*. For *Construction* class, the IoU of Dense-UNet reached to 68.54%, while for *Herbage* class, the IoU reached to 86.85%.

Fig. 7 shows visual comparison of the segmentation results obtained by flat experiments using U-Net with different backbones. It can be seen that different backbones were able to achieve good results for the classes of *Construction*, *Nonconstruction*, and *Herbaceous*. For *Water surface* and *Broad-leaved forest* classes, the segmentation performance of MobileNet and EfficientNet was the poorest. When using ResNet50 and DenseNet-121 as backbones, the identification for *Water surface* significantly improved, and the boundaries of different classes were much clear. As the label respects some generalization constraints, some results can stitch more to the raw image while exhibiting slightly lower performance metrics as being less coherent with the OCS-GE targeted product and its geometric specifications.

B. Land Cover Semantic Segmentation With the Proposed HierU-Net Method

The proposed HierU-Net is implemented and the three backbones including VGG19, ResNet50, and DenseNet-121 that performed well in the flat experiment are considered. The performance on coarse-level and fine-level classes on the validation dataset is shown in Table III. OA and mIoU values for coarse-level land cover types are in the range of 88.81% – 89.96% and 77.27% – 78.27%, respectively. For

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TABLE II
PERFORMANCE COMPARISON OF U-NET WITH DIFFERENT BACKBONES FOR EACH CLASS IN THE FINE CATAGORY

Model	Construction	Non-construction	Mineral Material	Water Surface	Broad Leaved Forest	Shrubbery	Herbage
U-Net	65.97	60.62	21.24	52.37	53.67	0.61	85.65
Mobile-UNet	51.66	46.77	4.43	24.07	47.03	0.32	80.73
Efficient-UNet	53.19	50.01	0.13	39.78	46.66	0.01	81.43
VGG-UNet	66.11	62.33	**23.22**	62.16	54.62	**19.60**	86.47
Res-UNet	67.28	61.21	10.93	63.59	54.90	6.07	86.17
Dense-UNet	**68.54**	**63.30**	11.79	**63.70**	**55.62**	0.54	**86.85**

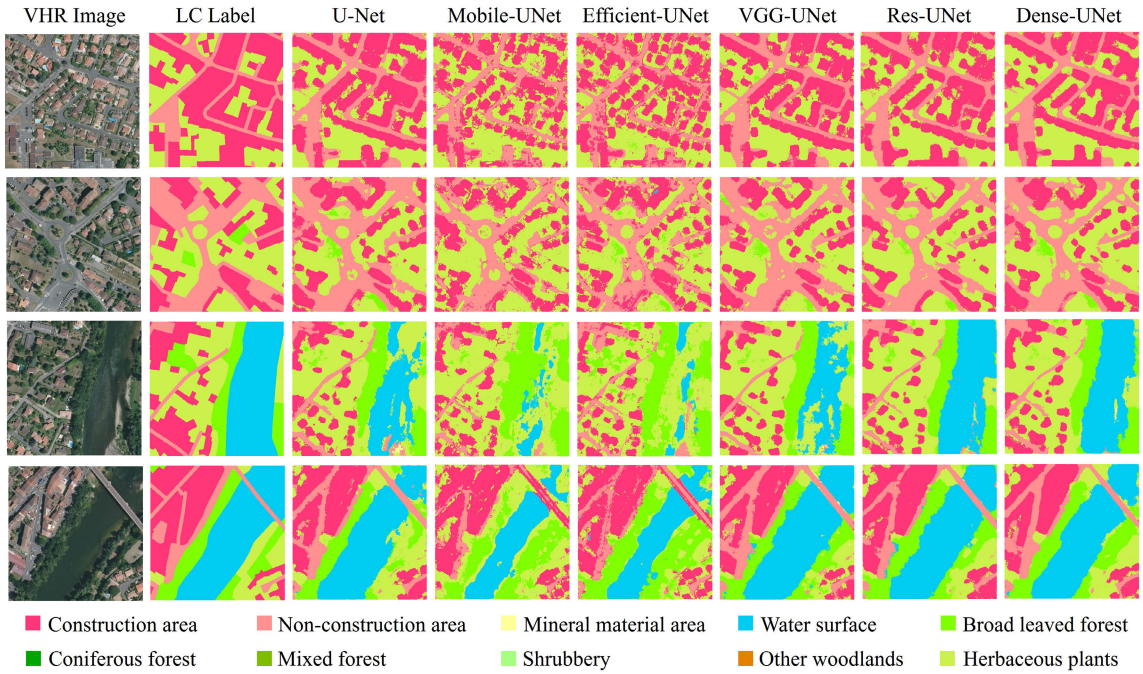


Fig. 7. Flat semantic segmentation results of input images for fine classes. From left to right are the VHR image, ground truth, and results from U-Net models with different backbones respectively.

TABLE III
PERFORMANCE COMPARISON OF HIERARCHICAL SEMANTIC SEGMENTATION USING DIFFERENT BACKBONES

Model	Level	OA	PA	UA	mIoU	FWIoU
HierU-VGGNet	Coarse	88.81	86.14	84.31	77.27	80.69
	Fine	84.58	46.59	71.87	38.96	74.92
HierU-ResNet	Coarse	89.59	86.65	85.59	77.90	81.26
	Fine	85.86	44.14	76.29	38.21	75.56
HierU-DenseNet	Coarse	**89.96**	**86.92**	**89.55**	**78.27**	**81.94**
	Fine	**87.18**	**49.90**	**76.62**	**42.71**	**77.76**

fine-level segmentation, OA and mIoU values are within the range of 84.58% – 87.18% and 38.21% – 42.71%, respectively. The results demonstrate that the model using DenseNet-121 as the encoder yields the most favorable segmentation for both coarse-level and fine-level land cover classes.

To validate the effectiveness of the proposed HierU-Net, we designed comparative experiments using additional semantic segmentation models, including SegFormer [53], SegNeXt [54], and a multitask semantic segmentation model based on U2Net and Histogram of oriented gradients (HOGs), referred as U2Net+HOGs [55]. The U2Net+HOGs employ an unsupervised approach to generate pseudo-labels for auxiliary tasks using HOGs. By combining the ground-truth semantic segmentation masks as the primary task with the pseudo-labels as the auxiliary task, the model collectively optimizes parameters of the deep network to minimize the losses of both tasks enhancing segmentation performance.

Per-class segmentation results in the fine category on the test dataset are summarized in Table IV. It is evident that SegNeXt has the fewest parameters and the lowest mIoU value as 26.77%. For SegFormer, the number of parameters is 84.60M achieving an unsatisfied result with a mIoU value of 26.87%. As a multitask model with ResNet50 and DenseNet-121 as backbones, HierU-Net outperforms U2Net+HOGs, achieving higher mIoU values by 0.87% and 1.59%, respectively. Notably, when DenseNet-121 is used as the backbone, the HierU-Net attains the highest IoU values in classes including *Construction*, *Mineral material*, and *Broad-leaved forest*. The

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TABLE IV
PERFORMANCE COMPARISON OF DIFFERENT MODELS FOR EACH CLASS IN THE FINE CATEGORY

Method	Backbone	Construction	Non-construction	Mineral Material	Water Surface	Broad Leaved Forest	Shrubbery	Herbage	mIoU	Params(M)	FPS
SegFormer-B5	MiT-B5	58.06	50.84	14.2	22.71	46.76	13.39	82.75	26.87	84.60	13.66
SegNeXt	MSCAN-B	57.47	52.07	11.29	16.18	44.55	5.59	80.52	26.77	33.61	17.40
U2Net+HOGs	-	68.06	**64.81**	26.87	69.26	56.49	20.61	87.05	39.32	44.14	10.98
HierU-Net	VGG-19	64.19	60.16	25.92	70.07	56.02	23.38	86.67	38.64	57.12	8.59
	ResNet50	68.71	63.19	25.53	**75.30**	56.89	**24.87**	**87.37**	40.19	287.23	7.34
	DenseNet-121	**69.69**	64.65	**30.61**	74.85	**57.40**	24.67	87.26	**40.91**	93.97	9.90

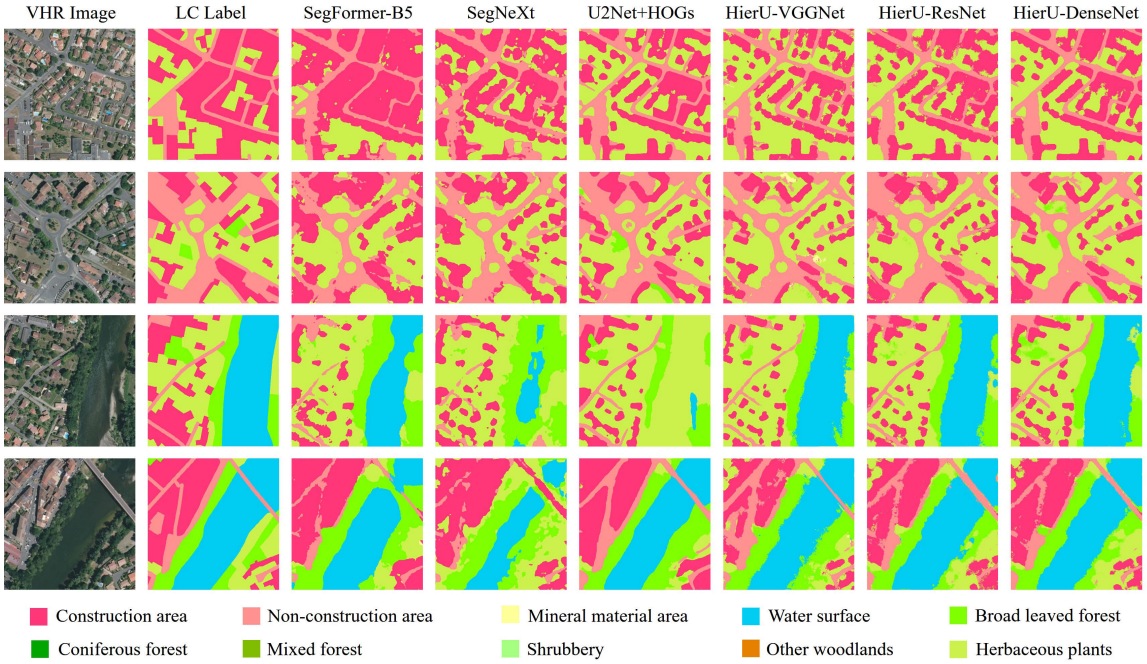


Fig. 8. Examples of semantic segmentation results. From left to right are the VHR image, ground truth, and results from SegFormer, SegNeXt, U2Net+HOGs, and HierU-Net using different backbones.

IoU for the *Herbage* reaches to 87.37%, while the lowest is observed in *Shrubbery*, ranging between 23.38% and 24.87%, still exhibiting significant improvement over flat methods. However, it is observed that HierU-Net introduces more parameters, resulting in increased GPU memory usage and a lower FPS value. To visually compare the performance of different models, examples are plotted in Fig. 8 along with the corresponding VHR imagery and land cover labels at the fine category.

The hierarchical segmentation model with DensNet-121 as the backbone is further considered for the test imagery. One of them is shown in Fig. 9 including the VHR imagery, predictions at two hierarchical levels, and the corresponding land cover labels. Overall, the model could correctly classify the majority of pixels in both coarse-level and fine-level classes. In Fig. 9(b), it can be seen that the class of *Mineral material* is underestimated, while the corresponding area is correctly identified as *Artificial surface* at the coarse-level segmentation. It demonstrated the challenge of fine-grained land cover mapping. The classes of *Mixed forest* and *Coniferous forest* that only constitute a small proportion and unevenly distributed in the datasets were not well learned by the model. Due to the similar appearance in the imagery, these two classes were prone to be misclassified as *Broad-leaved forest* in the fine-level prediction, as demonstrated in Fig. 9(a). Moreover, the ground truth data indicates that there is a small river in the middle of the imagery labeled as *Nature surface*. Corresponding pixels were identified as *Woody vegetation* at the coarse category and *Broad-leaved forest* at the fine category, as shown in Fig. 9(c). Indeed, it appears masked by trees on the imagery.

Furthermore, the geometry of OCS-GE is supposed to respect some generalization constraints defined formally in the specifications of the product for cartographic purposes. Thus, results reached by different models are generally a compromise between this generalization level and what actually appears in VHR imagery. This also explains why some results can visually seem better, stitching more to the raw images, while exhibiting slightly lower performance metrics. Indeed, they are less coherent with the OCS-GE targeted product and less fit with its geometric specifications. For instance, this can be observed in Fig. 8, where a unique *Construction* polygon in OCS-GE is classified into several individual buildings separated by vegetation, corresponding to the actual situation in VHR images. Using the coarse-level result as an input of Network-2 helps to maintain this product generalization level.

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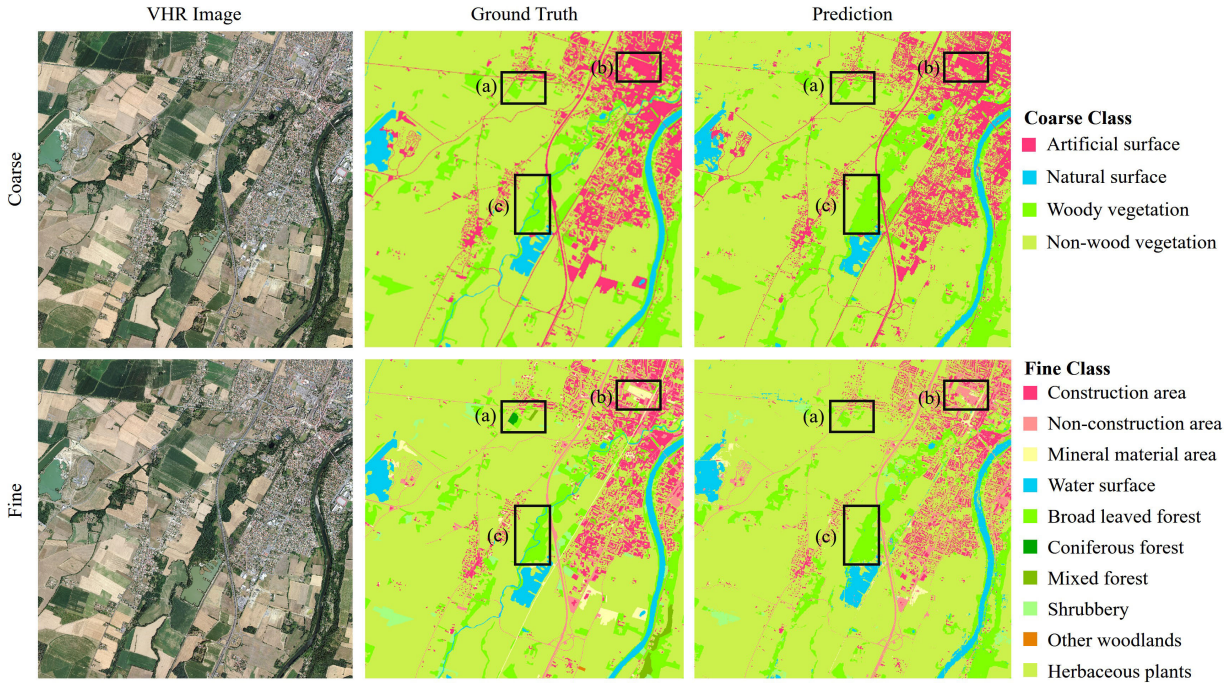


Fig. 9. Segmentation results of the HierU-Net model on the test set for coarse-level and fine-level classes.

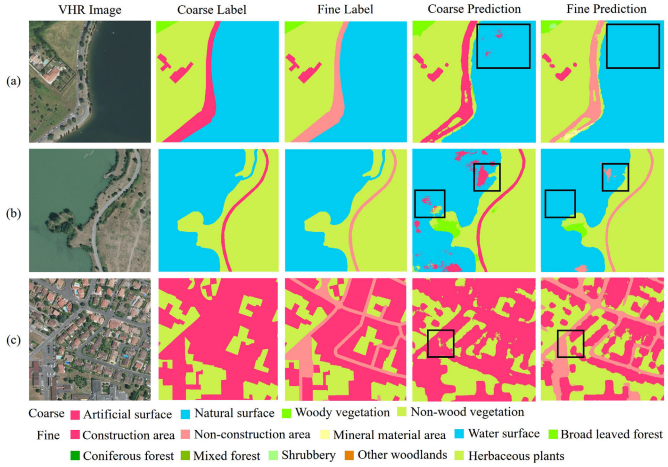


Fig. 10. Comparison of the coarse-level and fine-level segmentation of VHR images using the HierU-Net model (a) and (b) located in rural areas, and (c) located in urban area.

C. Comparison of Flat and Hierarchical Semantic Segmentation

The U-Net and proposed HierU-Net are able to segment the VHR imagery at the fine category. It can be seen that the hierarchical segmentation model achieved slightly better results on classes of *Construction* and *Nonconstruction*, and significantly improved the performance of *Mineral material*, *Water surface*, and *Shrubbery*. The IoU values of *Mineral material* increased from 11.79% to 30.61%. For *Shrubbery*, the IoU values increased from 0.54% to 24.67%.

To further analyze the segmentation consistency between the coarse and fine categories for the hierarchical semantic segmentation, examples are shown in Fig. 10 including the prediction at both categories and the corresponding ground-truth land cover labels. In general, the segmentation results keep a strong consistency between these two hierarchical categories. Most of the pixels identified as *Water surface* at the fine prediction correspond to the class *Nature surface* at the coarse category, and areas classified as *Construction* and *Nonconstruction* basically belong to *Artificial surface*. From Fig. 10(a) and Fig. 10(b), it can be seen that some areas in *Water surface* were misclassified as *Artificial surface* at the coarse category, but were correctly identified as *Water surface* at the fine category. The errors occurring at the coarse-level segmentation are not certain to be propagated to the fine-level segmentation.

D. Visual Interpretation via Saliency Maps

To understand what representations the hierarchical model has learned from the input VHR image, we visualized the saliency maps generated from class activation maps (CAMs) through a process that involves global average pooling on the feature map of the final convolutional layer [56]. Subsequently, weights at various positions are multiplied within the feature map to generate activation maps pertaining to specific classes. Saliency maps are overlaid on the original image, enabling us to gain insights into the network's decision-making and the features influencing the network's segmentation results. Fig. 11 illustrates some examples of saliency maps generated from CAM for the fine-level segmentation. Pixels with higher thermal values reflect the greater likelihood belonging to the corresponding land cover class. It can be seen that the saliency map basically localizes the fine-level land cover target class well.

E. Ablation Study

An ablation study is conducted to investigate the effectiveness of components in the proposed HierU-Net including

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TABLE V
ABLATION STUDY ON DIFFERENT COMPONENTS

Baseline	Coarse input	ASPP	MT loss	Level	OA	PA	UA	mIoU	FWIoU
✓				Coarse	89.29	82.85	88.39	75.32	80.69
				Fine	84.86	43.79	73.58	37.97	74.55
✓		✓	✓	Coarse	89.56	84.73	88.63	76.44	81.49
				Fine	86.65	47.68	75.27	40.72	76.75
✓	✓		✓	Coarse	89.37	84.67	88.90	76.41	81.51
				Fine	86.50	47.82	75.23	40.93	76.84
✓	✓	✓		Coarse	89.42	84.54	88.79	76.35	81.01
				Fine	86.38	48.20	74.98	40.84	76.65
✓	✓	✓	✓	Coarse	**89.96**	**86.92**	**89.55**	**78.27**	**81.94**
				Fine	**87.18**	**49.90**	**76.62**	**42.71**	**77.76**

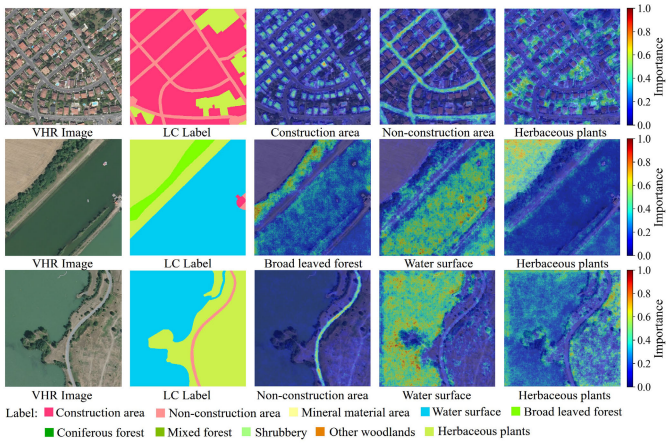


Fig. 11. Saliency maps generated from Class Activate Maps of classes at the fine-level category.

ASPP module, coarse input, and multitask loss function (MT loss) weighted by homoscedastic uncertainty during training. The baseline represents a model composed by two U-Nets with DenseNet-121 serving as the backbone. The second U-Net does not use the coarse result as inputs while both U-Nets are trained jointly. To make comparison with MT loss, losses for both tasks were computed using the cross-entropy loss function with equal weights. The experimental results on test dataset are presented in Table V.

When coarse-level segmentation results are not used as inputs for fine-level segmentation, the mIoU for the fine level is 40.72%, representing a decrease of 1.99% compared to the scenario where all components are implemented. Interestingly, there is also a decrease in the coarse-level segmentation results, with mIoU decreasing from 78.27% to 76.44%. Thus, coarse-level segmentation results contribute to the segmentation performance at both levels.

Upon removing the ASPP module from the network, the mIoU and OA values for fine categories decrease by 1.78% and 0.68% respectively, indicating that the addition of the ASPP module enhances the model's performance. Furthermore, replacing the employed MT loss with standard cross-entropy loss reduces the fine-level mIoU and OA by 1.87% and 0.80%, respectively. The optimal results are achieved when all components are implemented. In comparison to the baseline, the mIoU values for both coarse and fine levels increase by 2.95% and 4.74%, respectively.

F. Experiments on the GID Dataset

To prove the feasibility of the proposed HierU-Net, we conduct experiments on the widely used public GID dataset and compare with other segmentation methods. The GID dataset comprises high-resolution Gaofen-2 (GF-2) satellite images with extensive geographic distribution. An expanded version of the dataset (GID-15) is provide by [57] refining the original 5 land cover categories into 15 categories. The dataset contains 25,200 patches of 512 × 512 pixels—we divide them into two subsets—with 20 160 patches used for training and 5040 patches for testing.

Table VI presents the per-class IoU results obtained by different methods on the GID-15 dataset, along with the mIoU on the test set. The HierU-Net with DenseNet-121 as the backbone achieves the highest mIoU of 53.93%. Compared to the flat U-Net method using the same DenseNet-121 backbone, HierU-Net improves mIoU by 2.91%. For SegFormer and SegNeXt, HierU-Net shows higher mIoU outperforming by 6.39% and 14.84% respectively. In comparison to the multitask U2Net+HOGs approach, HierU-Net achieves a higher mIoU by 1.45%. Analyzing per-class IoU, HierU-Net demonstrates strong performance on frequent classes like *Urban residential*, *Paddy field*, and *Irrigated land*. For minority classes like *Garden land* and *Artificial meadow*, HierU-Net is surpassed by other methods.

Segmentation examples illustrated in Fig. 12 highlight HierU-Net's ability to preserve the geometric features compared to other methods. In contrast, approaches like SegFormer and SegNeXt produce internal holes and misclassified salt-and-pepper noises. The U2Net+HOGs method also shows inconsistent segmentation of *Traffic land* and *Paddy field*. Overall, the proposed HierU-Net model generates accurate segmentation maps, validating its effect in utilizing the tree-like structure of nomenclatures for fine-grained land cover mapping.

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TABLE VI
PERFORMANCE COMPARISON OF DIFFERENT MODELS FOR EACH CLASS IN THE FINE CATEGORY ON THE GID-15 DATASET

Method	Backbone	Ind. land	Urb. resid.	Rur. resid.	Traff. land	P. field	Irr. land	Dry cropl.	Garden
U-Net	DenseNet-121	**55.21**	65.45	**65.89**	50.76	82.31	65.61	64.74	0
SegFormer-B5	MiT-B5	48.46	58.45	48.93	15.99	72.94	57.31	**68.60**	3.90
SegNeXt	MSCAN-B	42.51	52.39	42.87	8.36	67.33	56.14	43.31	1.21
U2Net+HOGs	-	46.11	58.27	53.28	42.14	70.12	45.46	56.57	**18.12**
HierU-Net	DenseNet-121	52.32	**66.18**	61.79	**51.41**	**82.94**	**66.38**	63.55	2.79
Method	Backbone	Arb. forest	Shr. land	Nat. mead.	Art. mead.	River	Lake	Pond	mIoU
U-Net	DenseNet-121	77.99	72.68	42.15	0	58.70	78.10	18.64	51.02
SegFormer-B5	MiT-B5	77.83	**81.36**	**46.37**	6.59	59.00	**84.73**	26.11	47.54
SegNeXt	MSCAN-B	63.15	55.93	26.49	3.06	48.33	57.01	11.66	39.09
U2Net+HOGs	-	68.52	64.04	17.69	**16.94**	**59.74**	77.81	**29.17**	52.48
HierU-Net	DenseNet-121	**79.03**	73.76	43.84	4.26	53.74	78.76	21.45	**53.93**

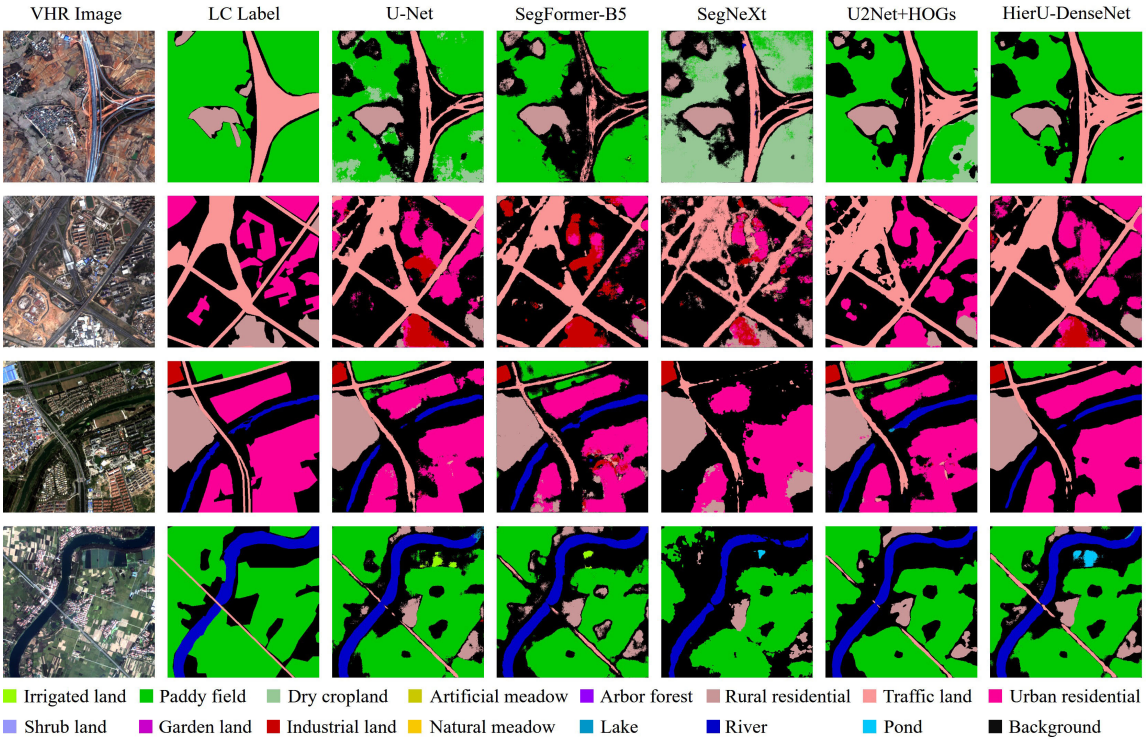


Fig. 12. Examples of semantic segmentation results on the GID-15 dataset. From left to right are the VHR image, ground truth, and results from SegFormer, SegNeXt, U2Net+HOGs, and HierU-Net.

V. CONCLUSION

In this article, a deep learning-based hierarchical semantic segmentation approach for multilevel land cover mapping has been introduced. First, a dual U-shaped network is proposed to learn the representation at coarse-level and fine-level land cover mapping simultaneously. Subsequently, the coarse-level segmentation results are ingested as inputs for fine-level segmentation, acting as soft constraints for fine-grained segmentation. Then, a joint loss function weighted by homoscedastic uncertainty is proposed to balance tasks at different levels. HierToulouse dataset is presented to promote the study of hierarchical semantic segmentation for land cover mapping.

The proposed HierU-Net is implemented on two datasets, including HierToulouse and GID-15. Experimental results demonstrate that HierU-Net can effectively achieve more accurate land cover segmentation compared to flat (U-Net, SegFormer, and SegNeXt) and multitask (U2Net+HOGs) methods. HierU-Net can be viewed as a multitask segmentation framework providing an intuitive yet effective way to distinguish coarse and fine categories simultaneously. It is flexible and could be extended to other segmentation modules.

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Another advantage is that the propagation of errors will not be certain by ingesting coarse-level information as soft constraints. The limitation is that the proposed method involves more parameters and has a lower FPS value. It can be alleviated by using shared weights between two U-shaped networks and fine-tuning the last one.

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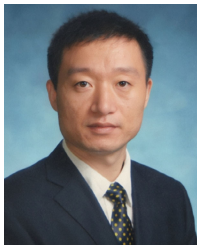
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